GSD RESEARCH MISSION PACKAGE — PART D

SO YOU WANT TO BE AN ENGINEER

NASA Engineering Deep Dive Template

Mechanical Engineering & Materials Science

The Math, the Metal, and the Mission

*Input: **{MISSION_NAME}** — Engineering Record from Parts A & B*

Part A (History) → Part B (Science) → Part C (Education) → **Part D** (Engineering) → Release (v1.50.**{SLUG}**.ENG)

CORE DISCIPLINES: Mechanical Engineering · Materials Science · Structural Analysis · Thermal Engineering · Engineering Mathematics
Systems Engineering · Manufacturing & Testing

Date: March 29, 2026 | Status: Template — Ready for Parameterization
Activation Profile: Fleet (16 roles) | Est. 7–10 context windows per mission run
Target: GSD College of Knowledge — Engineering & Physics Departments

Pipeline Architecture: Parts A+B → Part D → ENG Release

```
FOUR-PART PIPELINE — ENGINEERING LAYER ==========================================
PART A: NASA MISSION HISTORY → Mission catalog: era, type, hardware
overview, inheritance graph
PART B: SINGLE-MISSION DEEP RESEARCH → science_resultsJSON + hardware_record :
"spacecraft_design" : mass, power, propulsion, comms, "launch_vehicle" : name, thrust, staging, propellants, "key_engineer
PART C: SCIENCE EDUCATION PACK → CK study guides; Try Sessions; Claude
teaching scripts → (Part D is parallel to Part C — both consume Part B
output)
PART D: ENGINEERING DEEP DIVE (THIS DOCUMENT — TEMPLATE) Input:
Part B hardware_record + PartAmissioncontextFocus : MechanicalEngineering +
MaterialsScienceEngineeringMathematicsunderpinningeachdiscipline"Soyouwanttobeanengineer"career/skillframi
Engineeringcurriculumpack—studyguides, mathworkbooks, designexercises, materialsdatasheets, engineeringnotebook
Claudeengineeringtutorscripts + CKEngineeringDepartmentcontributions
RELEASE: gsd-skill-creator v1.50.MISSION_SLUG.ENG→CollegeofKnowledge :
EngineeringDepartment→GSD − OS : Engineeringdisciplinebrowser→Careerpathintegration :
"Howtobecomea[role]"entries
```

What distinguishes Part D from Parts B and C:

- Part B asks *what did we discover?* — Part D asks *what did we have to invent to get there?*
- Part C teaches the science to students — Part D teaches the engineering and mathematics to aspiring engineers
- Part D surfaces the real engineering math: stress tensors, phase diagrams, Fourier heat equations, fracture mechanics, rocket equation derivations — not simplified analogies, but the actual tools of the discipline
- Part D explicitly connects each engineering challenge to the career paths of the engineers who solved it: structural engineer, materials scientist, thermal engineer, systems engineer, manufacturing engineer

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STAGE 1 — VISION DOCUMENT

1 NASA Engineering Deep Dive — Vision Guide

Template Version: v1.0 (Part D of NASA Mission Engineering series)

Input Mission: **{MISSION_NAME}** (from Part B hardware_record)

Mission Slug: **{MISSION_SLUG}**

Part B Release: v1.50.**{MISSION_SLUG}** (must be complete before Part D)

EDU Release: v1.50.**{MISSION_SLUG}**.EDU (Part C; Part D runs in parallel)

ENG Release Target: gsd-skill-creator v1.50.**{MISSION_SLUG}**.ENG

Primary Disciplines: Mechanical Engineering, Materials Science, Engineering Mathematics

1.1 Vision

Every NASA mission that left the atmosphere had to solve engineering problems that had never been solved before. Not just technical challenges in the abstract — real, specific, career-defining problems that required people who understood the mathematics of stress, heat, vibration, and material failure to invent new answers under extreme time pressure with no margin for error.

The Apollo heat shield could not fail. The Cassini RTG could not leak. The Hubble primary mirror had to be ground to tolerances measured in nanometers. The James Webb Space Telescope's beryllium mirror segments had to stay aligned to within a fraction of a wavelength of light while cooling to 40 Kelvin in deep space. None of this happened because engineers had a hunch. It happened because engineers knew the mathematics of their disciplines deeply enough to design with confidence, test to failure deliberately, and solve novel problems systematically.

Part D exists to make that engineering knowledge accessible and teachable. Not as a watered-down survey, but as a serious engagement with the actual disciplines: structural analysis, materials selection, thermal engineering, mechanism design, manufacturing metrology, and the mathematics that makes all of them precise. The framing is deliberately aspirational — *So you want to be an engineer* — because the right response to encountering the engineering behind a NASA mission is not just admiration. It is the recognition that these are learnable skills with known curricula, established career paths, and communities of practice that a motivated person can enter.

For each mission in the Part A catalog, Part D answers: what engineering problems had to be solved, what mathematical tools were required, what materials were selected and why, what mechanisms had to work without failure in an environment where nobody could go fix them — and what a student needs to learn to eventually solve problems like these themselves.

1.2 Problem Statement

P1. Engineering knowledge locked in technical reports: The engineering innovations of NASA missions live in internal design documents, AIAA conference papers, and NASA Technical Reports Server entries that are rarely synthesized into teachable curriculum. Part D unlocks this knowledge for learners.

P2. Mathematics presented without engineering context: Students learn stress-strain relationships, Fourier heat equations, and eigenvalue problems in isolation. Part D situates these mathematical frameworks in the real engineering decisions of **{MISSION_NAME}** — the actual numbers, the actual constraints, the actual consequences of getting the math wrong.

P3. Materials science invisible in mission histories: Part B and Part C cover science discoveries and teaching curriculum. Neither treats materials science as a primary subject. Yet the choice of material for a spacecraft component is often the most consequential engineering decision made. Part D makes materials selection and materials science central.

P4. Mechanical engineering fragmented across disciplines: Structural analysis, thermal engineering, vibration, fluid mechanics, and mechanism design are taught as separate courses. In a spacecraft, they are coupled — a thermal load causes structural stress; a vibration mode drives fatigue; a mechanism's mass budget constrains the structural margins. Part D shows those couplings as they actually appear in **{MISSION_NAME}**.

P5. Career path invisible from science-first narrative: Parts B and C emphasize science and discovery. Part D adds the engineering career perspective — what role a structural engineer, materials scientist, thermal analyst, or systems engineer played in **{MISSION_NAME}**, and what education and experience path leads there.

1.3 Core Concept

Model: Each NASA mission as an engineering school — a set of real, unsolved problems that forced the development or application of specific engineering disciplines, mathematical frameworks, and materials capabilities, all of which remain teachable and career-relevant today.

1.3.1 The Five Engineering Pillars (Universal Across All Missions)

```
FIVE ENGINEERING PILLARS — APPLICABLE TO EVERY MISSION
==========================================================
PILLAR 1: STRUCTURAL MECHANICS The spacecraft must survive launch loads,
thermal cycling, and operational loads without yielding, fracturing, or
fatiguing. Math: Stress tensors, Mohr's circle, beam bending, buckling
Finite element analysis fundamentals, safety factors Materials interface:
Yield strength, ultimate strength, fatigue life
PILLAR 2: THERMAL ENGINEERING The spacecraft must maintain components
within operating temperature bounds across environments ranging from
direct sunlight to deep space. Math: Fourier conduction equation,
Stefan-Boltzmann radiation, Thermal network analysis, transient heat
```

```
equations Materials interface: Thermal conductivity, emissivity, α/ε ratio
PILLAR 3: MATERIALS SCIENCE  SELECTION Every component is made of
something. The choice determines mass, strength, thermal behavior,
and mission viability. Math: Crystal structure geometry, phase diagram
reading, Ashby charts, specific strength calculations Mechanics interface:
Modulus, CTE, fracture toughness, creep
PILLAR 4: MECHANISM DESIGN  DYNAMICS Deployable structures, gimbal
systems, articulating arms, valve assemblies — mechanisms that must work
in vacuum, at cryogenic temperature, after months of launch vibration.
Math: Kinematics equations, gear analysis, vibration modes, Natural
frequency, damping ratios Materials interface: Tribology, cold welding,
lubricant selection
PILLAR 5: SYSTEMS ENGINEERING  MARGINS The whole must work even when
individual components are at the edge of their specifications. Mass,
power, and reliability budgets govern every decision. Math: Link budgets,
mass fraction equations, reliability FMEA, Probability of success
calculations Materials interface: Mass properties, outgassing, radiation
tolerance
```

1.3.2 The Engineering Mathematics Hierarchy

Part D traces the mathematical lineage underlying each engineering decision in **{MISSION_NAME}**. Every engineering pillar has foundational mathematics that must be understood before the engineering application is meaningful:

Engineering Pillar	Foundational Math	Mission Application
Structural Mechanics	Linear algebra, calculus, differential equations	Stress at a fastener joint; panel buckling load; fatigue life calculation
Thermal Engineering	Partial differential equations, numerical methods	Heat rejection through radiator panels; transient cool-down time
Materials Science	Chemistry, crystallography, thermodynamics	Phase diagram reading; alloy selection for cryogenic use
Mechanism Design	Trigonometry, dynamics, vibration theory	Natural frequency of a deployable boom; gear ratio for antenna drive
Systems Engineering	Probability, statistics, optimization	Mass budget allocation; reliability target from FMEA

1.3.3 Module Architecture

```
SIX-MODULE ENGINEERING CURRICULUM ARCHITECTURE ==========================================
MODULE 1: ENGINEERING REQUIREMENTS  HERITAGE What problems
HAD to be solved for MISSION_NAME?→Requirementsflow −
downfrommissionobjectivestoengineeringspecs→Heritage : whatwasinheritedvs.whathadtobeinvented→Tradestudyexa
massvs.strengthvs.cost→Theroleofsafetyfactorsinaerospaceengineeringculture
```

```
MODULE 2: MECHANICAL ENGINEERING DEEP DIVE Structural, dynamic, and
thermal-mechanical analysis → Primary structural analysis: loads, stress,
safety margins → Vibration environment: launch acoustic + random vibration
→ Mechanism design: deployables, actuators, bearings → Thermal-structural
coupling: CTE mismatch stress calculation → Math: full treatment of stress
analysis for key components
MODULE 3: MATERIALS SCIENCE  SELECTION Every material
choice for MISSION_NAME—justified→Primarystructuralmaterials :
whythisalloy, notthatone→Thermalmaterials : coatings, MLI, ablatives, radiators→Mechanismsmaterials :
lubricants, bearings, coldwelding→Novelmaterials : whatwasnewaboutMISSION_NAME'smaterialchoices→Math :
Ashbychartmethodology, specificstrength, CTEcalculations
MODULE 4: ENGINEERING MATHEMATICS WORKBOOK The actual math — worked
through for MISSION_NAME→Rocketequation : derivationandapplication→Structural :
stressatacriticaljoint(realmissiongeometry)→Thermal : steady —
stateheatbalanceforamissioncomponent→Materials : materialselectionviapropertyindices→Mechanics :
naturalfrequencyofadeployablecomponent
MODULE 5: MANUFACTURING, TESTING  VERIFICATION How MISSION_NAMEcomponentsweremadeandproven—
protovs.qualificationvs.acceptance→Environmentaltesting : thermal —
vacuum, vibration, acoustics→Metrologyandinspection : howtoleranceswereverified→Theculturefaerospacequalityand
MODULE 6: ENGINEERING CAREER PATHS  LEGACY How to become the engineer who
solves these problems → Role profiles: structural, thermal, materials,
mechanisms, systems → Education pathway: degrees, skills, tools,
certifications → Innovations from MISSION_NAMEinusetoday→Engineeringlegacy :
whatsubsequentmissionsinherited→Resources : textbooks, softwaretools, professionalsocieties
```

1.4 Research Modules

1.4.1 Module 1: Engineering Requirements & Heritage

Traces the requirements chain for **{MISSION_NAME}** from mission-level objectives ("survive transit to Saturn") down through system-level requirements ("spacecraft must survive 3,600 kg launch mass in 6g axial load environment") to component specifications ("primary structure shall not yield at 125% design limit load"). Documents what engineering was directly inherited from predecessor missions (heritage design) and what had to be newly developed (new development). For each major new development, documents the trade study that selected the solution approach.

Key engineering questions: What was the single most demanding structural requirement of **{MISSION_NAME}**? What heritage components carried over from predecessors? What new development items drove the schedule and technical risk?

1.4.2 Module 2: Mechanical Engineering Deep Dive

The central technical module. Covers the structural analysis of the primary load-bearing elements, the vibration and acoustic environment and how components were designed to survive it, the mechanism design of critical deployable or articulating systems, and the thermal-structural coupling analysis where temperature gradients create stress. Includes real numbers from **{MISSION_NAME}**'s engineering record wherever available in open literature.

Structural focus: Primary structure, propulsion support structure, instrument mounting. Documents wall thickness decisions, material selection, and calculated safety margins.

Dynamics focus: Launch vibration environment (random, sinusoidal, acoustic), first natural frequency requirements, modal analysis approach, vibration isolation systems.

Mechanisms focus: Solar array deployment, antenna pointing mechanisms, instrument scan platforms, valve assemblies. Documents actuator selection, bearing specification, and mechanism-level testing.

1.4.3 Module 3: Materials Science & Selection

The dedicated materials module — the element most often absent from mission-level narratives. Covers every significant material choice in **{MISSION_NAME}**: the primary structural alloy or composite, the thermal protection material (if applicable), the optical coatings, the lubricants, the adhesives, the fastener materials, and any novel or mission-unique materials that were developed or first applied on this mission.

For each material, documents: what property drove its selection (specific strength? thermal conductivity? emissivity? radiation tolerance?), what the second-best candidate was, what testing was required to qualify it, and whether the material was subsequently adopted by other missions.

Special attention: Novel or first-use materials. What did **{MISSION_NAME}** contribute to the materials library that aerospace engineers now use routinely?

1.4.4 Module 4: Engineering Mathematics Workbook

The mathematics module — the element that transforms Part D from a history into a curriculum. Works through, with full derivations appropriate to the difficulty level, the key equations behind **{MISSION_NAME}**'s engineering decisions. Every worked problem uses real parameters from the mission where available in open sources, or representative parameters from the mission class where exact values are not public.

The workbook addresses: the Tsiolkovsky rocket equation (every mission), a stress analysis problem keyed to **{MISSION_NAME}**'s primary structure, a thermal analysis problem from the mission's thermal design, a material selection problem using the Ashby methodology, and a dynamics problem from the mission's vibration environment. Problems are presented at three levels: conceptual (algebra), working engineer (calculus), and advanced (numerical/FEM approach).

1.4.5 Module 5: Manufacturing, Testing & Verification

Covers the fabrication and verification of **{MISSION_NAME}**'s key components: what manufacturing processes were used (machining, composite layup, electron beam welding, precision grinding), what test philosophy governed the program (protoflight, qualification/flight build, etc.), and what the key environmental tests looked like (thermal-vacuum cycling, vibration table, acoustic chamber, mass properties).

Special attention to the metrology and quality culture of aerospace manufacturing: how tolerances are measured, how traceability is maintained, what "acceptance test" means versus "qualification test," and why the culture of aerospace quality is itself a learnable and transferable engineering skill.

1.4.6 Module 6: Engineering Career Paths & Legacy

The "so you want to be an engineer" payoff module. Profiles the engineering roles that were central to **{MISSION_NAME}**: structural analyst, thermal engineer, materials engineer, mechanisms engineer, systems engineer, manufacturing engineer, test engineer. For each role: what a typical day looks like, what education is needed (BS/MS/PhD?), what tools are used (NASTRAN, ANSYS, NX, MATLAB, Python), what professional societies are relevant, and what a career progression from entry level to lead engineer looks like.

Documents the specific engineering innovations from **{MISSION_NAME}** that persisted into subsequent missions or commercial applications — the engineering legacy that is parallel to the scientific legacy covered in Parts B and C.

1.5 Chipset Configuration (Template)

```
chipset:  Part D: Engineering-specific role configuration
agnus:  Engineering Orchestration model: opus role: FLIGHT + CRAFT-ENG
responsibility: > Cross-pillar engineering synthesis; trade study
evaluation; math correctness judgment; materials selection rationale;
career path narrative; engineering culture voice
denise:  Document Assembly — Technical Writing model: sonnet role:
EXEC-STRUCT + EXEC-MAT responsibility: > Module 2 structural/dynamic
content; Module 3 materials content; engineering specifications tables;
materials data sheets; technical drawing descriptions (ASCII format)
paula:  Math Workbook + Exercises model: sonnet role: EXEC-MATH +
CRAFT-PROB responsibility: > Module 4 worked examples; problem set
generation; equation derivations; numerical examples with mission
parameters; three-level problem scaffolding (conceptual/working/advanced)
gary:  Verification + Requirements Architecture model: sonnet role:
CRAFT-REQT + VERIFY responsibility: > Module 1 requirements flow-down;
heritage vs. new-development classification; engineering claim
verification against Part B; safety factor documentation; NASA standards
cross-reference
68000:  Haiku Scaffolding model: haiku role: BUDGET + LOG + PLAN
responsibility: > Engineering parameter schema; CK YAML scaffolding;
career path template; token tracking; release packaging
model_allocation : opus : 30 sonnet : 60 haiku : 10
```

1.6 Scope Boundaries

1.6.1 In Scope (every Part D run)

- Mechanical engineering analysis (structural, thermal, vibration, mechanisms) specific to **{MISSION_NAME}**
- Materials science: selection rationale, property data, novel materials, qualification testing
- Engineering mathematics: fully worked examples using mission parameters
- Manufacturing and test philosophy for key components
- Systems engineering: requirements, margins, trade studies, FMEA

- Engineering career path profiles for roles central to **{MISSION_NAME}**
- CK Engineering Department contributions (study guides, problem sets, data sheets)
- Cross-mission engineering inheritance links (from Part A graph)
- Engineering legacy: innovations that persisted to subsequent missions
- Release artifact: v1.50.**{MISSION_SLUG}**.ENG

1.6.2 Out of Scope

- Detailed electrical/electronic engineering (avionics, power systems, comms hardware) — separate future Part E
- Software engineering, flight software, guidance algorithms — separate future Part F
- Science instrument internal engineering (covered in Part B)
- Propulsion chemistry (covered in Part B chemistry context)
- Classified or ITAR-controlled technical details
- Component-level drawings or proprietary specifications
- Science results and discoveries (Part B)
- Teaching curriculum for non-engineers (Part C)

1.6.3 Engineering Evidence Calibration

Evidence Level	Mission Examples	Open Literature	Calibration
Rich	Apollo, Hubble, Cassini, ISS	Extensive NTRS, AIAA papers	Full treatment; real numbers
Standard	Most flagship missions	Some NTRS; mission reports	Full treatment; some representative numbers
Sparse	Early Mercury, some probes	Limited open documentation	Document what's available; flag gaps
Active	JWST, Perseverance, Parker	Growing literature	Current as of run date; flag preliminary
Proprietary risk	All missions	Some specs ITAR-controlled	Representative industry parameters used

1.7 Success Criteria

SC1. Module 1 establishes the top-level engineering requirements for **{MISSION_NAME}** and classifies each major subsystem as heritage or new development.

SC2. Module 2 provides structural analysis of at least one primary structural element: loads, material, calculated stress, safety factor, and design decision rationale.

SC3. Module 2 identifies the launch vibration environment and describes how at least one component was designed or tested to survive it.

SC4. Module 2 covers at least one mechanism (deployable, pointing, or valve) with actuator type, bearing specification, and heritage or development history.

SC5. Module 3 provides a materials selection rationale for at least three distinct components, with candidate comparison and selection basis.

SC6. Module 3 identifies at least one novel or first-use material in **{MISSION_NAME}** and documents its properties and qualification approach.

SC7. Module 4 delivers at least five fully worked engineering problems using **{MISSION_NAME}** parameters: rocket equation, stress analysis, thermal analysis, material selection calculation, and one dynamics problem.

SC8. Module 4 presents each worked problem at three levels: conceptual, working engineer, and advanced.

SC9. Module 5 covers the manufacturing process for at least one key component and the test philosophy for the mission (qualification/flight or protoflight approach).

SC10. Module 6 profiles at least four engineering career roles central to **{MISSION_NAME}** with education requirements, tools, and professional societies.

SC11. All engineering claims cite Part B or open-source technical literature (NASA NTRS, AIAA papers, mission reports); no engineering claims introduced without a source.

SC12. Release artifact correctly versioned as v1.50.**{MISSION_SLUG}**.ENG; deposited to correct output path.

1.8 The Through-Line

The hardest problems in aerospace engineering are not solved by people who are smart. They are solved by people who know their mathematics deeply enough to reason about the problem before building anything, and know their materials well enough to choose the right medium for the solution before the first drawing is made. Intuition is earned through mastery of fundamentals.

The Saturn V F-1 engine was the most powerful liquid-fueled rocket engine ever flown — and it suffered catastrophic combustion instability during early testing that threatened the Apollo program. The solution was not found by guessing. It was found by Rocketdyne engineers who understood the acoustics of a combustion chamber well enough to design baffles that changed the acoustic mode structure of the combustion zone. They knew the mathematics of acoustic resonance. They knew the materials that could survive the resulting thermal and structural loads. They solved it by knowing things.

Part D is a doorway into knowing those things — not at the level of a casual observer,

but at the level of someone who wants to actually do this work. Every module in this template is a chapter of the engineering education that the people who built **{MISSION_NAME}** had completed before they were given the problem to solve. The math is here. The materials are here. The problems are here. The career path is here. What comes next is up to the student.

STAGE 2 — RESEARCH REFERENCE

2 NASA Engineering Deep Dive — Research Reference

2.1 How to Use This Reference

Part D agents consume Part B's `hardware_record` JSON as their primary input. Secondary input is the Part A mission record for heritage/successor context. Unlike Parts B and C, Part D agents also conduct targeted research in the NASA Technical Reports Server (NTRS), AIAA digital library, and mission-specific technical documentation to source engineering details not captured in Part B.

2.2 Input Bundle Schema

```
INPUT BUNDLE SCHEMA — PART D ENGINEERING RECORD ==========================================
From: Part B release + Part A catalog
"mission_name" : "string","mission_slug" : "string","mission_era" : "string","mission_type" :
"string",
"hardware_record" : "spacecraft" : "name" : "string", //e.g."CassiniOrbiter""dry_mass_kg" : number, "launch_mass_kg"
"predecessors": ["string"], // from Part A — for heritage documentation
"successors": ["string"], // for engineering legacy section "part_bsources"  :
["string"]//PartBbibliography—first − tiersources
```

2.3 Source Hierarchy

Tier	Source Type	Examples for Engineering Content
T1	NASA Technical Reports Server	ntrs.nasa.gov — design documents, test reports, lessons learned
T2	AIAA conference and journal papers	AIAA SciTech, Journal of Spacecraft and Rockets
T3	Mission-specific JPL/GSFC publications	Mission System Engineering documents, publicly released
T4	NASA NTIS / mission press kits	Press kit engineering sections; fact sheets with mass/power data
T5	Peer-reviewed materials journals	Acta Materialia, Composites Science and Technology
T6	Engineering textbook cross-reference	Shigley's, Incropera, Ashby — for math framework only

ITAR and proprietary boundary: If a specific engineering parameter is not in T1–T5 sources, use class-representative values from analogous missions. Flag all representative values explicitly. Never estimate classified performance data.

2.4 Research Protocol by Module

2.4.1 Module 1: Requirements & Heritage Research

Query Template	Target
`{MISSION_SLUG} system requirements specification`	Top-level engineering requirements
`{MISSION_NAME} mass budget structural requirements`	Mass and structural load requirements
`{MISSION_NAME} heritage design from {PREDECESSOR}`	What was directly inherited
`{MISSION_NAME} new development technology risk`	Novel engineering items
`{MISSION_NAME} trade study selection rationale`	Design decision documentation
`NASA systems engineering handbook spacecraft`	NASA SE methodology framework

2.4.2 Module 2: Mechanical Engineering Research

Query Template	Target
`{MISSION_NAME} structural analysis primary structure`	Main load paths; stress analysis results
`{MISSION_NAME} launch vibration acoustic test`	Environmental test conditions and survival
`{MISSION_NAME} deployable mechanism solar array`	Mechanism design; actuator details
`{MISSION_NAME} thermal control design radiator`	Thermal architecture; radiator sizing
`{MISSION_NAME} NTRS structural design report`	Technical report with engineering detail
`{MISSION_NAME} natural frequency modal survey`	Vibration analysis; first mode frequency

2.4.3 Module 3: Materials Science Research

Query Template	Target
`{MISSION_NAME} material selection {COMPONENT}`	Per-component material with justification
`{MISSION_NAME} novel material first use spacecraft`	Novel materials first applied on this mission
`{MISSION_NAME} thermal protection ablative coating`	Heat shield or coating material details
`{MISSION_NAME} composite carbon fiber structure`	Composite usage and layup approach
`aerospace material {MATERIAL_NAME} properties`	T5/T6 property data for identified materials
`{MISSION_NAME} material qualification test NTRS`	How novel materials were proven

2.4.4 Module 4: Engineering Mathematics Research

Module 4 is produced from first principles (the mathematics is well-established) using **{MISSION_NAME}** parameters sourced from T1–T4. The workbook agent (EXEC-MATH) derives the mathematics independently; the mission parameters are the inputs, not the derivations. Cross-reference:

- Rocket equation: launch vehicle specific impulse, propellant mass fraction from mission data
- Stress problem: structural load, material yield strength, cross-section geometry from Part B hardware record
- Thermal problem: solar flux at mission orbit, surface area, absorptivity/emissivity from thermal control description
- Material selection: properties of at least three candidate materials from T5/T6; Ashby index calculation
- Dynamics problem: component geometry and material for natural frequency estimate

2.4.5 Module 5: Manufacturing & Test Research

Query Template	Target
`{MISSION_NAME} manufacturing process fabrication`	How key components were made
`{MISSION_NAME} thermal vacuum test qualification`	Environmental test approach
`{MISSION_NAME} protoflight qualification test plan`	Test philosophy documentation
`{MISSION_NAME} mass properties measurement`	Mass measurement and uncertainty
`aerospace spacecraft acceptance test procedures`	General aerospace test culture
`{MISSION_NAME} anomaly test failure lessons learned`	Test failures and resolution

2.4.6 Module 6: Career Paths Research

Query Template	Target
`NASA structural engineer career path education`	Structural engineer role at NASA
`aerospace materials engineer skills tools`	Materials engineer role profile
`AIAA thermal engineer spacecraft career`	Thermal engineer profile
`systems engineer NASA JPL requirements skills`	Systems engineer role profile
`{MISSION_NAME} engineering innovation commercial`	Legacy applications outside NASA
`spacecraft mechanisms engineer tribology career`	Mechanisms engineer profile

2.5 Engineering Mathematics Framework Reference

The following equations are universally relevant across all missions. Module 4 always includes these, populated with mission-specific parameters:

Equation	Domain	Mission Application
$\Delta v = I_{sp} g_0 \ln(m_0/m_f)$	Rocket equation (Tsiolkovsky)	Propellant mass budget; mission Δv trades
$\sigma = F/A$ (simple); $\sigma = Mc/I$ (bending)	Structural mechanics	Primary structure stress at design limit load
$\nabla^2 T = \frac{1}{\alpha}\frac{\partial T}{\partial t}$	Fourier heat equation	Transient thermal response of spacecraft panel
$q_{rad} = \varepsilon \sigma T^4$	Stefan-Boltzmann radiation	Radiator sizing for heat rejection
$f_n = \frac{1}{2\pi}\sqrt{k/m}$	Natural frequency (SDOF)	First mode requirement for deployable structure
$\sigma_{y,specific} = \sigma_y/\rho$	Specific strength	Material selection for mass-critical structure
$\Delta T_{max} = Q/(UA)$	Thermal network (conductance)	Conductance path sizing for thermal interface
$K_{Ic} = \sigma\sqrt{\pi a} \cdot Y$	Fracture mechanics	Flaw tolerance in structural material

2.6 Materials Property Reference Schema

Every material identified in Module 3 is documented in a standard data sheet format:

```
MATERIALS DATA SHEET SCHEMA ============================
material_name : "string"//e.g.,"6061 − T6Aluminum"material_class :
"string"//Metal|Ceramic|Polymer|Composite|Coatingmission_application :
"string"//WhereusedonMISSION_NAME
mechanical_properties : density_kg_m3 : numberyoungs_modulus_GPa : numberyield_strength_MPa :
number_ultimate_strength_MPa : number_fracture_toughness_MPa_sqrt_m : number_fatigue_limit_MPa :
number|null
thermal_properties : thermal_conductivity_W_mK : numberspecific_heat_J_kgK : numberCTE_per_K :
number//coefficientofthermalexpansionmax_service_temp_K : numbermin_service_temp_K : number
space_environment_properties : radiation_tolerance : "string"//TIDratingor"excellent/good/limited"outgassing_CVCM
number|null//condensablevolatilescold_welding_risk : "string"//"high/medium/low"formechanismsUV_stability :
"string"
selection_basis : why_selected : "string"candidates_considered : ["string"]key_trade : "string"
qualification_status : first_use_on : "string"//MISSION_NAMEorearliermissionheritage_uses :
["string"]//subsequentmissionsusingthismaterial
```

2.7 Safety and Technical Accuracy Rules

Rule	Type	Application
No ITAR data	ABSOLUTE	If a parameter is classified or export-controlled, use representative class data; flag explicitly
Math correctness	BLOCK	All equations checked by FLIGHT (Opus); no incorrect derivations; units explicitly stated throughout
No invented properties	BLOCK	All material properties sourced from T5/T6; no estimated or interpolated values without flagging
Safety factor honesty	GATE	Document actual safety factors if in open literature; use “typical aerospace” range if not
Representative parameters flagged	GATE	Any number that is representative (not from **{MISSION_NAME}** directly) marked † with note
Career claim accuracy	ANNOTATE	Salary ranges, program durations, and job titles sourced from BLS or institutional data

STAGE 3 — MISSION PACKAGE

3 Milestone Specification

3.1 Mission Objective

Produce a complete College of Knowledge engineering curriculum pack for NASA mission **{MISSION_NAME}**, covering the mechanical engineering, materials science, and engineering mathematics required to understand and build upon the engineering achievements of the mission. Deliver structured study guides, worked mathematics, design exercises, materials data sheets, engineering notebooks, Claude engineering tutor scripts, and four engineering career path profiles. Release as v1.50.**{MISSION_SLUG}**.ENG to the GSD College of Knowledge Engineering Department.

3.2 Architecture Overview

```
WAVE EXECUTION ARCHITECTURE — MISSION_NAMEENGINEERINGPACK ====================
[WAVE 0: ENGINEERING INVENTORY + PARAMETER INJECTION] Ingest Part B
hardware_record; classifyengineeringchallengesbypillar; identifynovelmaterials; extractloadenvironment.(Haiku+
CRAFT−REQT)|
                |
                |[W1A:MECHENG][W1B:MATERIALS]Module2Module3Structural+MaterialsdataThermal+sheets;selectionMechanismsrationale;(So
```

3.3 Deliverables

#	Deliverable	Agent	Acceptance Criteria
1	Engineering challenge inventory (Module 1)	CRAFT-REQT	All challenges from Part B hardware_record classified; heritage/new-dev labelled
2	Requirements flow-down (Module 1)	CRAFT-REQT	Top-level → subsystem → component chain documented for primary structure
3	Structural analysis study guide (Module 2)	EXEC-STRUCT	Loads, material, stress calc, safety factor for at least one element
4	Vibration/dynamics study (Module 2)	EXEC-STRUCT	Launch environment; at least one component modal analysis
5	Mechanisms study guide (Module 2)	EXEC-STRUCT	At least one mechanism with actuator, bearing, heritage
6	Materials data sheets (Module 3)	EXEC-MAT	Full schema for ≥3 components; novel material identified
7	Materials selection rationale (Module 3)	EXEC-MAT	Candidate comparison for at least 2 selections; Ashby index noted
8	Math workbook — 5 problems × 3 levels (Module 4)	EXEC-MATH	All equations correct; units stated; mission parameters used
9	Manufacturing & test guide (Module 5)	EXEC-STRUCT	One component process description; test philosophy; key environmental test
10	Engineering career profiles × 4 roles (Module 6)	CRAFT-ENG	Education, tools, societies, progression per role
11	Engineering legacy analysis (Module 6)	CRAFT-ENG	≥2 innovations named; successor mission usage documented
12	Claude engineering tutor scripts	CRAFT-ENG (Opus)	Canonical explanations; misconceptions; engineering-appropriate boundary handling
13	CK Engineering Dept YAML entries	LOG	Schema validates; all modules linked
14	Release artifact v1.50.**{MISSION_SLUG**	LOG	Correct version; correct output path

3.4 Component Breakdown

Component	Dependencie	Model	Est. Tokens
W0: Engineering inventory	Part B YAML	Haiku	8K
W1A: Structural analysis guide	W0	Sonnet	28K
W1A: Vibration/dynamics guide	W0	Sonnet	22K
W1A: Mechanisms guide	W0	Sonnet	20K
W1B: Materials data sheets ($\geq$3)	W0	Sonnet	30K
W1B: Materials selection rationale	W0	Sonnet	18K
W2A: Math workbook (5 problems $\times$ 3)	W0	Sonnet+	45K
W2B: Requirements flow-down (Mod 1)	W0	Sonnet	18K
W2B: Manufacturing & test guide (Mod 5)	W1A	Sonnet	20K
W3: Career profiles $\times$ 4 roles	W1A,B,2A,B	Sonnet	24K
W3: Engineering legacy analysis	W1A,B	Opus	18K
W3: Claude tutor scripts	W1–W2	Opus	36K
W4: CK YAML + packaging	W3	Haiku	6K
Contingency (10%)	—	Mixed	30K
	Total per mission run:		**~323K**

3.5 Model Assignment Rationale

Model	Share	Rationale
Opus	30%	Math correctness verification (critical); engineering legacy judgment; Claude tutor script authorship (tone of a seasoned mentor engineer); trade study evaluation quality
Sonnet	60%	Study guide production for all modules; materials data sheets; worked math problems (with Opus verification); manufacturing guides; career path content
Haiku	10%	Engineering inventory schema; YAML scaffolding; release packaging; token tracking

4 Wave Execution Plan

4.1 Wave 0: Engineering Inventory

Task	Agent	Output
Ingest Part B hardware_record	BUDGET	Validated parameter bundle; pillar classification per challenge
Classify each challenge by pillar	CRAFT-REQT	Structural / Thermal / Materials / Mechanisms / Systems
Flag novel materials for Module 3	CRAFT-REQT	Novel-material list with first-use flag
Extract math parameter values	EXEC-MATH	Seed values for Module 4 workbook problems
Initialize CK YAML scaffolds	LOG	One engineering entry YAML per module

Wave 0 Go Condition: All five engineering pillars have at least one challenge assigned; at least three materials identified; math seed parameters available for all five problem types. BLOCK if Part B hardware_record is empty or missing key fields.

4.2 Waves 1 & 2: Parallel Production

Track A — Mechanical Engineering (Modules 2 + 5) runs in parallel with **Track B — Materials Science (Module 3)**:

Track	Key Deliverable Requirements
W1A: Structural	Primary structural member: material, geometry (approximate), design load, stress calculation, safety factor, design decision; sourced to T1–T4 or flagged representative
W1A: Vibration	Launch environment description (axial g, random vibration g_{rms}, acoustic dB); first natural frequency requirement; how at least one component was designed/tested to survive
W1A: Mechanisms	Named mechanism from Part B; actuator type; bearing type and lubricant (including cold-welding risk in vacuum); functional test approach
W1B: Data sheets	Full schema data sheet for $\geq$3 components using identified materials; all properties sourced; representative values flagged †
W1B: Selection	Candidate comparison for $\geq$2 material selections; Ashby performance index stated; why winner beat candidates

Wave 2A: Mathematics Workbook runs after W0 (seed parameters available) in parallel with W2B:

Each problem appears in a `mathbox` environment. All five problems are worked at three levels. Opus verifies all equations before Wave 3 proceeds.

Problem	Math Domain	Levels
P1: Rocket equation	Tsiolkovsky / exponential	Conceptual: plug-in; Working: Δv budget; Advanced: multi-stage optimization
P2: Structural stress	Mechanics of materials	Conceptual: axial stress $\sigma = F/A$; Working: bending + safety factor; Advanced: von Mises + margin
P3: Thermal balance	Radiation / conduction	Conceptual: equilibrium temperature; Working: radiator sizing; Advanced: transient cool-down
P4: Material selection	Ashby methodology	Conceptual: specific strength ranking; Working: performance index calculation; Advanced: multi-attribute decision
P5: Natural frequency	Dynamics (SDOF)	Conceptual: spring-mass analogy; Working: $f_n = \frac{1}{2\pi}\sqrt{k/m}$; Advanced: beam mode shape

4.3 Wave 3: Synthesis — Career Paths & Claude Tutor Scripts

Wave 3 is the Opus synthesis wave. CRAFT-ENG reads all W1/W2 content and produces:

Synthesis Task	Description
Career profile × 4 roles	Structural engineer, materials engineer, thermal engineer, and one mission-specific role (e.g., mechanisms engineer for missions with complex deployables). Each profile: education path, tools, professional societies, day-in-the-life, progression to lead
Engineering legacy analysis	Name ≥2 specific innovations from **{MISSION_NAME}** traceable to subsequent missions or commercial applications; write the causal paragraph for each
Claude tutor scripts	Canonical engineering explanations per pillar; engineering misconception map; "if student asks X, suggest Y" patterns for engineering reasoning; escalation from conceptual to working level; boundary handling for classified/ITAR questions
Math verification pass	Opus reviews all five Module 4 problems for mathematical correctness, unit consistency, and physical reasonableness of results
Cross-mission engineering links	From Part A inheritance graph: what did predecessors contribute? What did **{MISSION_NAME}** contribute that appears in successors?

Wave 3 Go Condition: All five math problems pass Opus verification; all four career profiles complete; legacy analysis identifies ≥2 innovations with causal explanation; CAPCOM notified.

4.4 Wave 4: CK Deposit & Release

Task	Agent	Gate
CK Engineering Dept YAML entries	LOG	Schema validates; all 14 deliverables linked
VERIFY spot-check (10 eng. claims)	VERIFY	All 10 traced to T1–T5; representative values flagged
Math correctness final check	VERIFY	2 problems re-derived independently; results match
Release packaging	LOG	All files at correct output path; version string correct
CAPCOM review	CAPCOM	Human approval before release tag

4.5 Cache Optimization Strategy

- **Pillar batching:** W1A processes all structural content before dynamics, then mechanisms. Each sub-track shares the structural material context, enabling cache reuse.
- **Math seed prefix:** All five W2A problems begin with the same mission-parameter table (mass, load environment, material properties). One cache hit covers all five problems.

- **Materials cross-reference:** If Module 3 material appears in Module 2 (same material, structural and thermal use), EXEC-STRUCT and EXEC-MAT share the data sheet rather than each generating separately.
- **Era batching:** Like Parts B and C, batch missions from the same era. Apollo-era missions share Aluminum 2024-T4, Inconel 718, and fiberglass MLI; these material context blocks are cached across all Apollo Part D runs.
- **Opus single-pass:** All Wave 3 Opus tasks (career profiles, tutor scripts, math verification, legacy analysis) run in a single Opus context window to maximize consistency.

4.6 Token Budget

Wave	Opus (K)	Sonnet (K)	Haiku (K)
Wave 0	—	6	2
Wave 1A (Mech Eng)	—	70	—
Wave 1B (Materials)	—	48	—
Wave 2A (Math workbook)	12	33	—
Wave 2B (Reqts + Mfg)	—	38	—
Wave 3 (Synthesis + Opus)	54	24	—
Wave 4 (CK + Release)	—	—	6
Contingency (10%)	7	22	1
Total	**73K**	**241K**	**9K**
		Grand Total:	**~323K**

4.7 Dependency Graph

```
DEPENDENCY GRAPH — MISSION_NAMEENGINEERINGPACK ==========================
PART B RELEASE + PART A CATALOG (external inputs) ↓ W0: ENGINEERING
INVENTORY (Haiku + CRAFT-REQT) ↓ +──────────────────────┬──────────────────+
| | | W1A: STRUCTURAL W1B: MATERIALS W2B: REQUIREMENTS GUIDE DATA
SHEETS + MFG GUIDE | (EXEC-STRUCT) (EXEC-MAT) (EXEC-STRUCT) | | |
W1A: VIBRATION/DYN W1B: SELECTION W2A: MATH WORKBOOK GUIDE RATIONALE
5 problems × 3 levels | (EXEC-STRUCT) (EXEC-MAT) (EXEC-MATH + Opus
verify) | | | W1A: MECHANISMS | | GUIDE | | (EXEC-STRUCT) | |
```

```
|

 |

  |[SYNC:AllW1+W2complete]|W3:SYNTHESIS(Opus−CRAFT−ENG)Careerprofiles×4+Legacy+Tutorscripts+Mathverificationpass|W4:CKYAML+VERIFY
```

5 Test Plan

5.1 Test Categories

Category	Count	Priority	On Fail	Description
Safety-Critical	5	P0	BLOCK	Math correctness, ITAR boundary, materials accuracy
Core Completeness	12	P1	BLOCK	One per success criterion
Engineering Quality	8	P2	REVISION	Technical accuracy; problem scaffolding; career realism
Integration	6	P2	REVISION	Cross-module consistency; pillar coupling
Edge Case	4	P3	ANNOTATE	Sparse data, classified parameters, active missions

5.2 Safety-Critical Tests (BLOCK on Failure)

Test ID	Verifies	Expected Result
SC-01	Math correctness (Opus verified)	All five Module 4 problems independently re-derived; results physically reasonable; no dimensional errors
SC-02	ITAR/proprietary boundary	No classified performance data present; all representative values flagged †; no export-controlled propellant specifics
SC-03	Materials properties sourced	All property values in data sheets trace to T5/T6 source; no invented or interpolated properties unflagged
SC-04	No new science claims	Part D introduces no science facts not in Part B; engineering context only
SC-05	Safety factor accuracy	Safety factors documented as actual (from open lit.) or “typical aerospace” range; never invented specific values

5.3 Engineering Quality Tests

Test ID	Quality Check	Acceptance Standard
EQ-01	Structural problem physical reasonableness	Calculated stress is < yield strength (otherwise design would have failed); safety factor is in credible aerospace range (1.25–2.0)
EQ-02	Thermal problem equilibrium check	Equilibrium temperature is within known operating range for that mission's orbit/environment
EQ-03	Material selection non-obvious	Selection rationale explains why the obvious choice (e.g., aluminum) was or was not used; candidate comparison is genuine
EQ-04	Three-level math scaffolding	Conceptual level requires only algebra; Working level introduces calculus or data lookup; Advanced level is genuinely open-ended
EQ-05	Career profile realism	Education requirements accurate for US aerospace industry; tool list reflects current industry practice; salary range sourced or omitted
EQ-06	Engineering legacy causal	Legacy statement is causal ("X enabled Y because Z") not correlational ("Y came after X")
EQ-07	Novel material genuinely novel	First-use claim verified against T1/T5 heritage dates; not routine
EQ-08	Mechanism vacuum consideration	Any mechanism guide discusses vacuum tribology and cold-welding risk; lubricant selection addressed

5.4 Integration Tests

Test ID	Cross-Module Check	Expected
IN-01	Module 2 ↔ Module 3 coupling	Material used in structural analysis matches material in data sheet; properties consistent
IN-02	Module 4 ↔ Module 3 coupling	Stress problem uses yield strength from Module 3 data sheet; thermal problem uses conductivity from data sheet
IN-03	Module 1 → Module 2 coverage	Every challenge classified "Structural" or "Mechanisms" in Module 1 appears in Module 2 study guide
IN-04	Module 6 → Module 2 legacy	Engineering legacy innovations are traceable to specific Module 2 or 3 engineering decisions
IN-05	Part A inheritance ↔ Module 1 heritage	Heritage items in Module 1 match predecessor missions listed in Part A inheritance graph
IN-06	Math workbook → Module 3 material selection	Module 4 Ashby problem uses materials from Module 3; winner matches Module 3 selection

5.5 Verification Matrix

SC#	Criterion	Test IDs	Status
SC1	Challenges classified; heritage/new-dev labelled	CF-01, IN-05	TBD
SC2	Structural analysis with stress and safety factor	CF-02, EQ-01, IN-01	TBD
SC3	Vibration environment; component survival design	CF-03, IN-03	TBD
SC4	Mechanism with actuator, bearing, heritage	CF-04, EQ-08	TBD
SC5	Materials data sheets ≥ 3; selection rationale	CF-05, SC-03, EQ-03	TBD
SC6	Novel material identified and qualified	CF-06, EQ-07	TBD
SC7	Math workbook: 5 problems, mission parameters	CF-07, SC-01, IN-02	TBD
SC8	Three-level math scaffolding per problem	CF-08, EQ-04	TBD
SC9	Manufacturing process + test philosophy	CF-09	TBD
SC10	Career profiles $\times 4$ with education and tools	CF-10, EQ-05	TBD
SC11	Engineering legacy ≥ 2 innovations, causal	CF-11, EQ-06, IN-04	TBD
SC12	Release versioned correctly; path correct	CF-12	TBD

6 Mission Crew Manifest

Role	Agent (PNW Naming)	Responsibility
FLIGHT	RAVEN	Mission direction; math verification oversight; trade study judgment; engineering culture voice
PLAN	SALMON	Wave decomposition; parallel track optimization; W2A math sequencing
CRAFT-ENG	OWL	Engineering synthesis; Claude tutor script authorship; career profile narrative; legacy causal analysis
CRAFT-REQT	EAGLE	Requirements flow-down; heritage classification; engineering challenge taxonomy; NASA SE methodology
EXEC-STRUCT	CEDAR	Modules 2+5: structural, vibration, mechanisms, manufacturing, test guides
EXEC-MAT	HEMLOCK	Module 3: materials data sheets; selection rationale; novel material documentation
EXEC-MATH	ALDER	Module 4: five worked problems × three levels; equation derivations; unit checking
VERIFY	HAWK	Source tracing; ITAR boundary check; math spot-check (2 independent re-derivations); properties sourcing
INTEG	HERON	Cross-module consistency; Module 2/3 property coupling; inheritance graph cross-reference
CAPCOM	FOXGLOVE	Human gate; revision loop; ITAR escalation if needed
BUDGET	WREN	Token tracking; era-batch optimization; math problem seed parameter management
LOG	MARTIN	CK YAML generation; release packaging; output path management; commit messages
TOPO	BEAR	Multi-track topology management; W1A/1B/2A/2B parallel coordination
ARCH	WOLF	Cross-pillar structural decisions; ensures thermal-structural coupling appears in both Mod 2 and 3
SECURE	CRANE	ITAR and proprietary boundary enforcement; flags any potential export-control issues before EXEC writes
RETRO	OSPREY	Post-wave analysis; documents lessons learned for Part D template improvement

7 Sample Deliverable: Worked Engineering Problem (Illustrative)

The following illustrates Module 4 output format for one problem at the Working Engineer level, using representative Cassini mission parameters.

Mission context: Cassini operated at Saturn, 9.5 AU from the Sun. The spacecraft used three Multi-Mission Radioisotope Thermoelectric Generators (MMRTGs) for power, each producing ~4.5 kW of thermal waste at beginning-of-mission. The spacecraft needed to reject excess heat via two ~3 m^2 radiator panels while maintaining electronics above 0°C.

Working Engineer Level:

Step 1: Solar flux at Saturn. $G_{Saturn} = G_{Earth}/d^2 = 1361\ \text{W/m}^2/(9.5)^2 \approx 15.1\ \text{W/m}^2$

Step 2: MMRTG total waste heat. $Q_{waste} = 3 \times 4{,}500\ \text{W} = 13{,}500\ \text{W}$

Step 3: Required radiator temperature (Stefan-Boltzmann). $Q = \varepsilon\sigma A T^4 \Rightarrow T = \left(\frac{Q}{\varepsilon\sigma A}\right)^{1/4}$

With $A = 6\ \text{m}^2$, $\varepsilon = 0.90$ (black anodize), $\sigma = 5.67 \times 10^{-8}\ \text{W/m}^2\text{K}^4$: $T = \left(\frac{13{,}500}{0.90 \times 5.67 \times 10^{-8} \times 6}\right)^{1/4} \approx 311\ \text{K} = 38°\text{C}\dagger$

Result: Radiators operating at ~38°C at beginning-of-mission, well above the 0°C electronics minimum. Thermal design margin is adequate. As RTGs decay over the 13-year mission, Q_{waste} decreases, requiring heater power to maintain minimum temperatures. This is why Cassini carried internal heaters.

$\dagger Representativecalculation; actualCassinithermaldesignincludesviewfactors, planetaryIR, conductionpaths, andmultinodemodel. SeeGanapathietal., AIAA1994-4014forCassinithermaldesignoverview.$

ENGINEERING NOTEBOOK: Engineering Notebook — The RTG Thermal Management Trade

The problem: RTGs produce far more heat than electrical power (efficiency ~6%). Most of that heat must go somewhere that doesn't overheat electronics.

Option A: Reject all waste heat to space via large radiators. Pro: Simple. Con: At Saturn, very cold environment risks overcooling during low-power phases; requires large, heavy radiator panels.

Option B: Use waste heat intentionally as spacecraft heater, reject the rest. This is what Cassini did. Heat pipes redistributed RTG waste heat to electronics bays. Louvers controlled heat rejection from radiators. The "bus" was kept warm at Saturn using the same heat that would be a cooling problem in Earth orbit.

Engineering lesson: In spacecraft thermal design, a "waste" is often a resource in disguise. The RTG heat that needed to be rejected in inner-solar-system orbit became the primary spacecraft heating system at Saturn. The engineer who recognized this reframing saved mass, simplified the design, and solved two problems with one thermal network.

Career note: Thermal engineers at JPL routinely use this kind of systems thinking. The tool is a thermal math model (SINDA/FLUINT or Thermal Desktop); the skill is knowing which heat loads to fight and which to embrace.

8 Execution Summary

Metric	Value
Activation Profile	Fleet (16 roles)
Est. Token Budget (median mission)	~323K
Est. Token Budget (flagship, Apollo/Cassini)	~450K
Model Allocation	Opus 30% / Sonnet 60% / Haiku 10%
Context Windows per Run	7–10
Wave Structure	0, 1A/1B/2A/2B (parallel), 3 (Opus synthesis), 4, 4B (CAPCOM)
Safety-Critical Gates	5 BLOCK tests (math correctness, ITAR, materials sourcing)
Engineering Quality Tests	8 REVISION gates
Math Problems	5 types × 3 levels = 15 worked problem variants
Career Profiles	4 roles per mission run
Output per Run	14 deliverables including study guides, data sheets, math workbook, career profiles, Claude tutor scripts
Release Version	v1.50.**{MISSION_SLUG}**.ENG
Output Path	`.planning/outputs/missions/`**`{MISSION_SLUG}`**`/eng/`
CK Department Target	Engineering + Physics (cross-links)
Parallel to	Part C (both consume Part B; run simultaneously)
Resumability	Tracked in `part-d-progress.json`

"Intuition is the residue of mastery. The structural engineer who looks at a fastener pattern and immediately knows it's undertorqued, or the materials scientist who holds a sample and suspects intergranular corrosion before the SEM confirms it — they are not guessing. They are doing rapidly what they once did slowly, with equations and reference books and test data. The fast path to good intuition is the slow path through the fundamentals."

Part D does not offer shortcuts. It offers the fundamentals of mechanical engineering and materials science as they were actually applied on **{MISSION_NAME}** — with real loads, real temperatures, real material properties, and real consequences. A student who works through the Module 4 problems with pencil and paper and then reads the engineering notebook for that same

mission component is doing, in miniature, the same work that the engineers at JPL or GSFC or MSFC did before the first drawing was released.

That is how you become an engineer. One correctly-solved problem at a time.